Roles & Permissions

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GCXONE uses Role-Based Access Control (RBAC) to manage what each user can do and what they can see.

How It Works

A role is the core unit of access control in GCXONE. Each role combines three elements:

Role Information & Users Name, description, default toggle, assigned users.

Module Privileges Which modules & capabilities users can access. Privileges determine WHAT a user can do.

Entity Assignment Which customers, sites, devices, and sensors users can access. Entity access determines WHERE a user can operate.

Entity assignments can be configured at two levels:

- Role-Level Entities: Entities assigned within the role itself, shared by all users in the role.
- User-Level Entities: Entities assigned to individual users through Edit Entity Access, allowing different users in the same role to access different customers or sites.

Key Distinction: Entity Access vs. Privileges

Privileges (configured in the role) determine WHAT actions a user can perform. Entity access determines WHICH customers, sites, devices, and sensors they can see. User-level customization is done through entity access only — there are no user-level privilege overrides.

Default Roles

GCXONE includes pre-configured roles to help you get started quickly. Each role comes with a defined set of module privileges. Entity access can be further customized per user.

Super Admin Full platform access with complete control over all modules, users, roles, and settings.

- Complete administrative access to all system features
- Can manage all users, roles, and permissions
- Full access to all entities
- Configure system-wide settings and audit logs
- One Super Admin per tenant. Cannot be deleted.

Admin Administrative access — can manage users, roles, configuration, and reports, but not billing.

- Manage users and role assignments
- Configure system settings
- Create and manage Entity Groups

- View audit logs and reports
- Cannot access billing
- Cannot delete critical system components

Operator Operational access — can view and manage configuration and reports but cannot delete or administer users/roles.

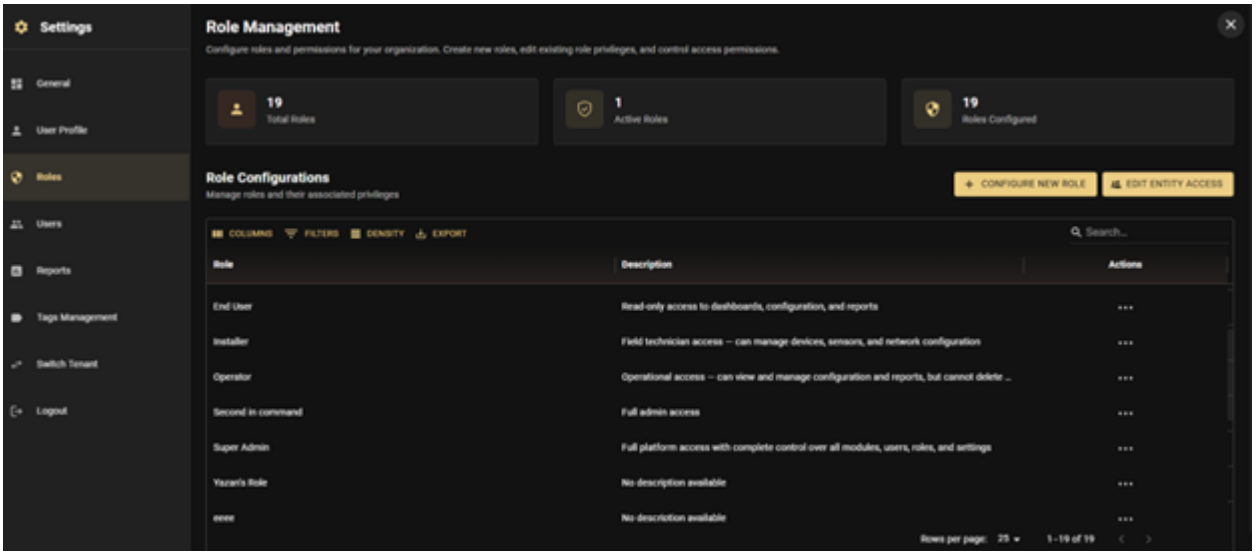
- Day-to-day monitoring and alarm processing
- View dashboards and configuration
- Generate and view reports
- Handle operational tasks
- Cannot delete system components
- Limited administrative capabilities

Installer Field technician access — can manage devices, sensors, and network configuration.

- Manage physical devices and sensors
- Configure network settings
- Install and set up hardware
- Troubleshoot device issues
- Cannot manage users or roles

End User Read-only access to dashboards, configuration, and reports.

- View dashboards and basic configuration
- Access basic reports
- View assigned cameras and devices
- Arm/disarm for assigned devices/sites
- No ability to change settings or administer the system



You can use these default roles as-is, customize them, or create new roles from scratch.

Core Concepts

Entity Hierarchy

GCXONE organizes entities in a hierarchical structure:

Service Provider (Tenant / Account) → Customer (Organization) → Site (Physical Location) → Device (Hardware) → Sensor (Camera / Detector)

Cascading selection: Selecting a customer automatically includes all sites, devices, and sensors beneath it. This ensures new entities are automatically included when using the Include Children toggle and simplifies configuration.

Service Provider (Tenant / Account)	Customer (Organization)	Site (Physical Location)	Device (Hardware)	Sensor (Camera / Detector)

Module Privileges

Privileges are organized by platform module. Each module has an Access Page toggle and granular capability settings. Privileges determine WHAT actions a user can perform across the platform.

- Dashboard: Access Page toggle, Accounts filter, Capability filter
- Management Insights: Enable Role, Edit Role, and additional analytics capabilities
- Configuration: Mobile Users, Roles, SubAnalytics, Health Check, Customer Groups, Sensor Table, Audits, Dealer Table, Service Provider Table, IoT, Group/Stage Table, Customer Group, Service Provider, Customer, Site, Configuration Dashboard
- Video Activity Search: Edit Saved, Use Filter, Live, Playback, Edit Filter, Alarm Filter
- Marketplace: Pulse View, Roles, Time Sync, Health Check, Timer Count, Zen Mode, Bulk Import
- Video Viewer: Video Controls, Capability toggles
- Alarm Manager: Video Controls, Capability toggles
- Map: Edit Map, Add Sensor, Add IO, Edit IO, Manage Review, Manage IO, Video Live, Live Stream View, Wall Monitor, View Dashboard

Each module has a Select All / Remove All option for quick bulk configuration, plus an Access Page toggle that enables or disables the entire module.

Entity Access: Three Modes

Entity access determines WHERE a user can operate — it controls which customers, sites, devices, and sensors are visible and accessible to the user.

No Entity Access Users have no entity access by default. Must be assigned per user via Edit Entity Access. Best for: Shared role with individual entity assignments.

Selected Entities Users are restricted to the specifically selected entities (with or without children). Best for: Specific sub-lists or tenants with identical access.

Full Access Users default to full access for all Customers, Sites, Devices, and Sensors. Best for: Admin or super-user roles needing unrestricted access.

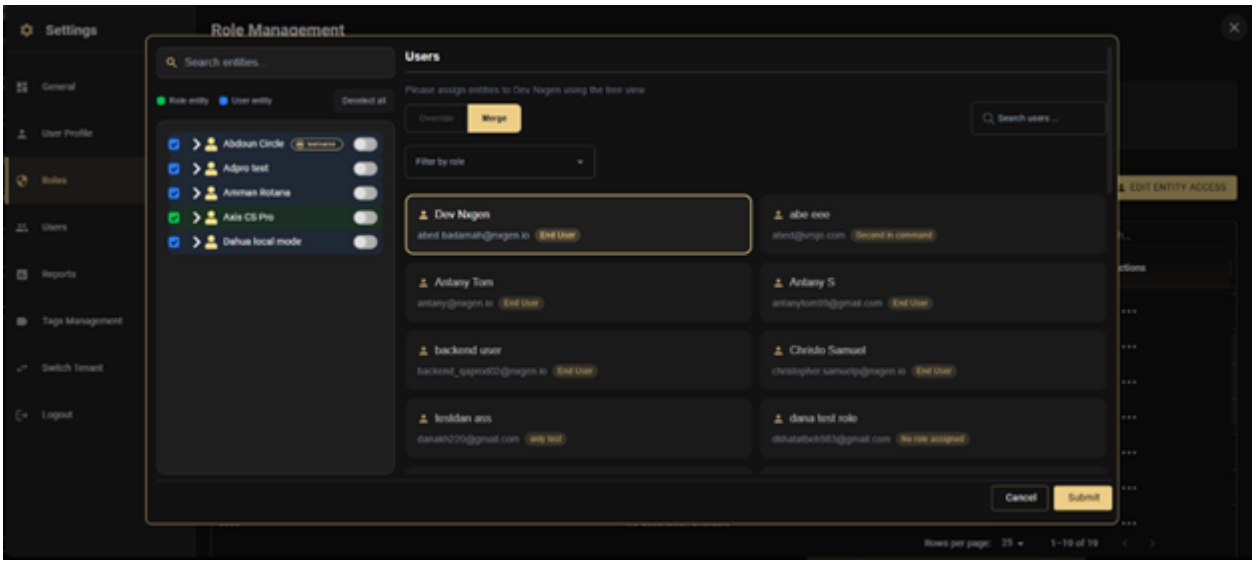
Regardless of mode, you can always customize the entity list for specific users through Edit Entity Access, overriding or extending the role-level assignments while keeping their role privileges intact.

The Include Children Toggle

When selecting entities in the tree view, each entity has an Include Children toggle.

Without Include Children: Only the manually selected entities are included. Future devices or sensors added under that site will NOT be included automatically — you would need to manually go back to Edit Entity Access and select the new device.

With Include Children: You only need to select the parent entity. All devices and sensors underneath are automatically included. Any future devices or sensors added under that site are automatically included — no manual updates needed.



Tip: Use Include Children whenever possible. It saves time and ensures new devices are automatically accessible to the right users. Only use manual selection when you specifically need to restrict access to a fixed set of entities.

Permission Resolution

When a user has multiple roles (direct assignments + group memberships), all privileges combine additively. Changes take effect within 5 seconds across web and mobile platforms.

Managing Roles

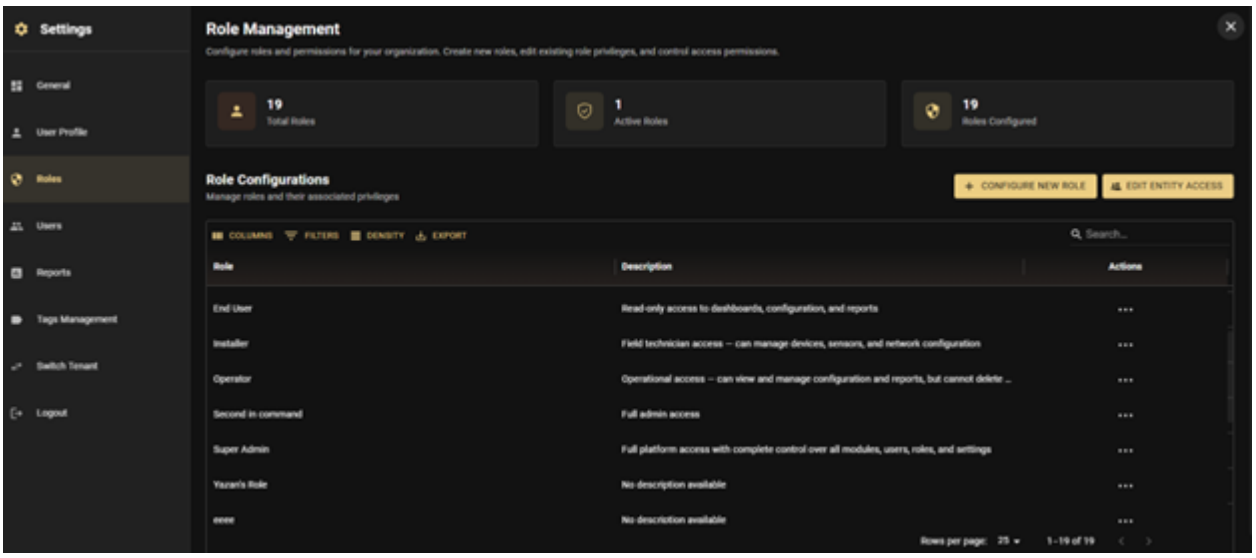
Role Management Overview

Navigation: Settings → Roles

The Role Management page displays all configured roles with their descriptions and provides two main action buttons:

- Configure New Role — Opens the guided role creation wizard

- Edit Entity Access — Opens the per-user entity access management interface



Creating a Role

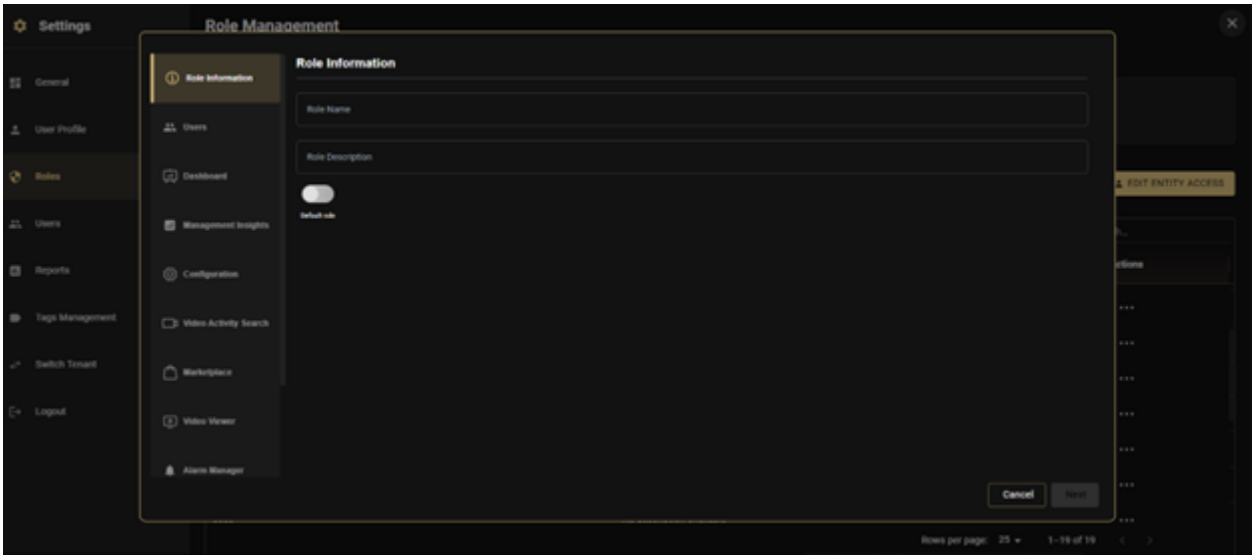
Navigation: Settings → Roles → Configure New Role

The role creation wizard guides you through a multi-step process:

1. Role Info — Name, description, default toggle
2. Users — Assign users to this role
3. Privileges — Module-by-module capability config
4. Entities — Entity access mode & selection

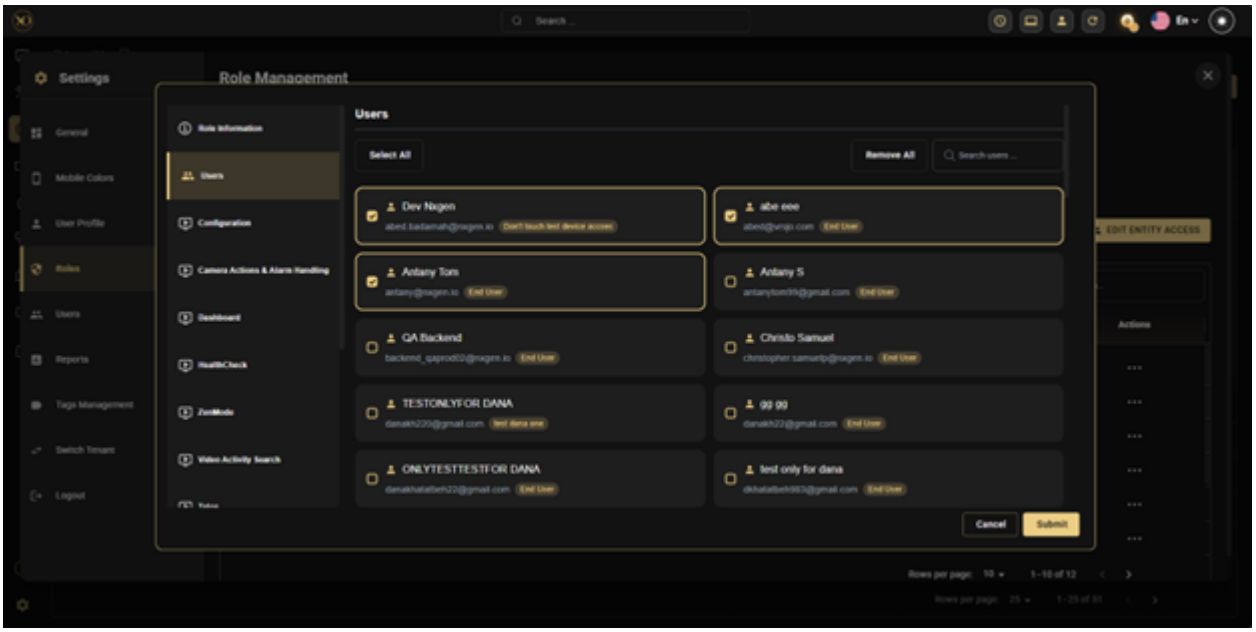
Step 1: Role Information

1. Role Name — A unique name identifying the role (e.g., "Regional Admin", "View-Only Operator")
2. Role Description — A brief description of the role's purpose
3. Default Role toggle — When enabled, this role will be automatically assigned to new users when they are invited to the platform. This is useful for base-level roles like "End User" that every new team member should start with.



Step 2: Users

Select which users should be assigned to this role. The Users step displays all available users in a searchable grid with their current role badges. Click on a user card to select them (highlighted with a gold border). Use Select All, Remove All, or Search Users for bulk management.

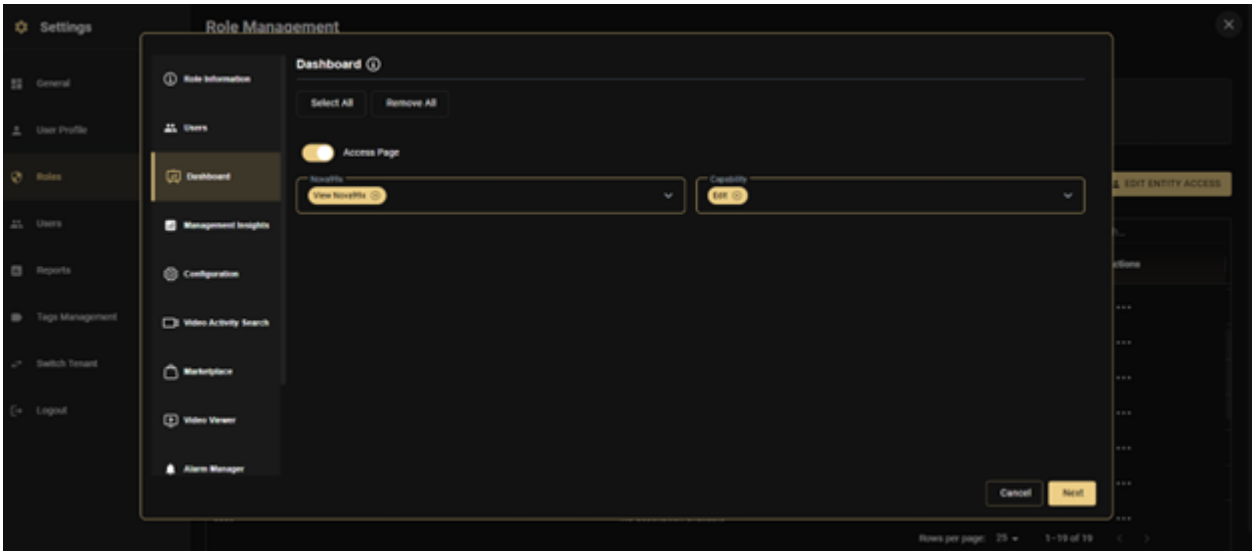


Step 3: Module Privileges

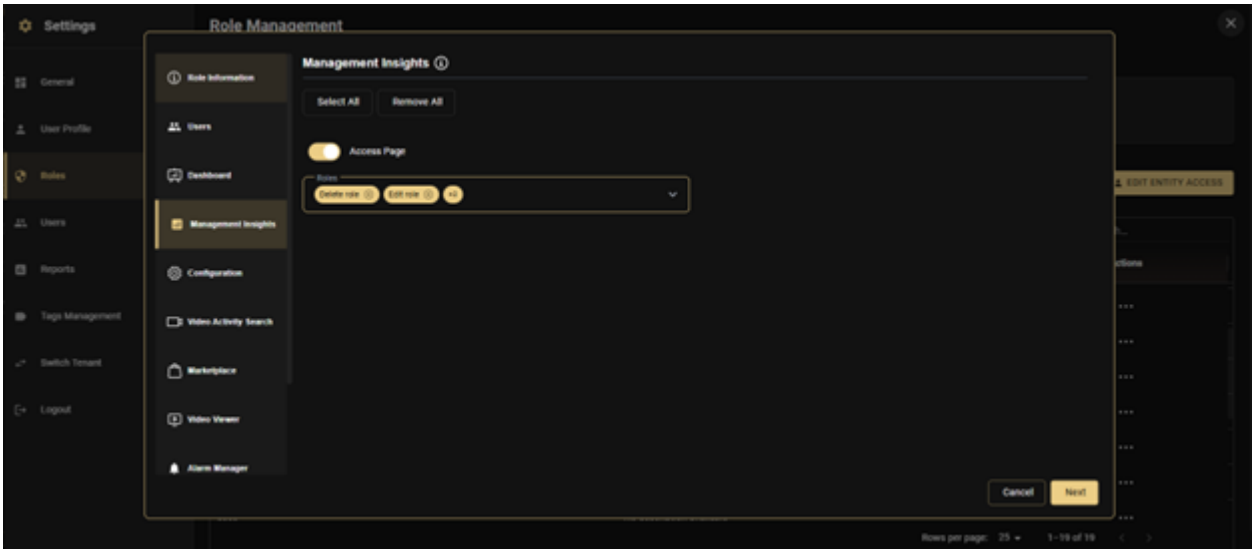
Configure access for each platform module. The wizard presents each module as a separate tab in the sidebar. For each module:

1. Toggle Access Page on/off to enable or disable the entire module
2. Use Select All / Remove All for quick bulk selection of capabilities
3. Select individual capabilities from the dropdown menus to fine-tune access

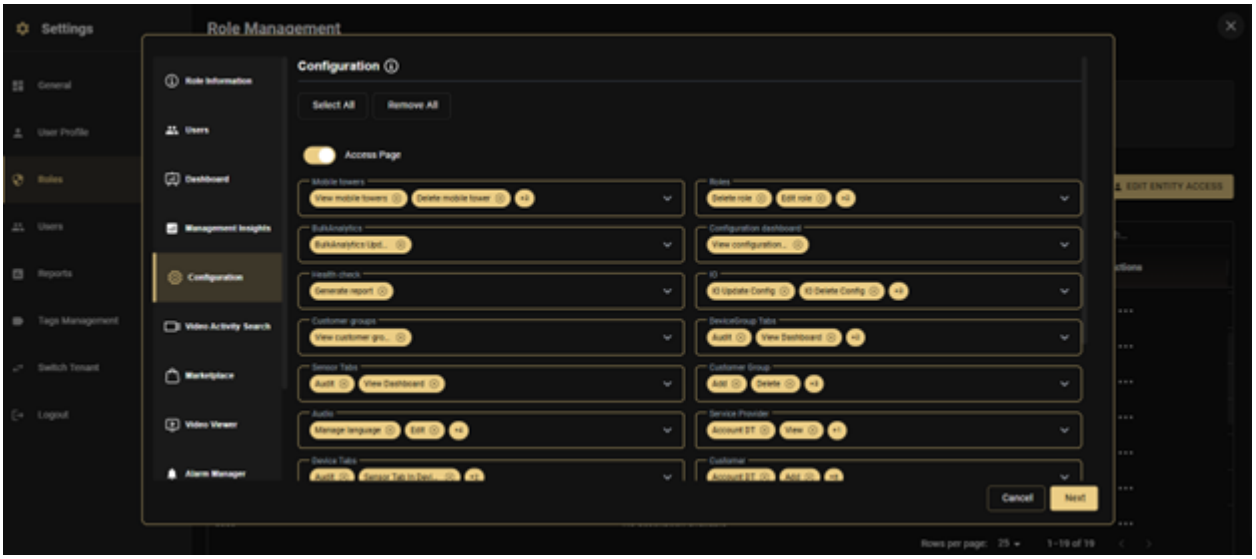
Dashboard:



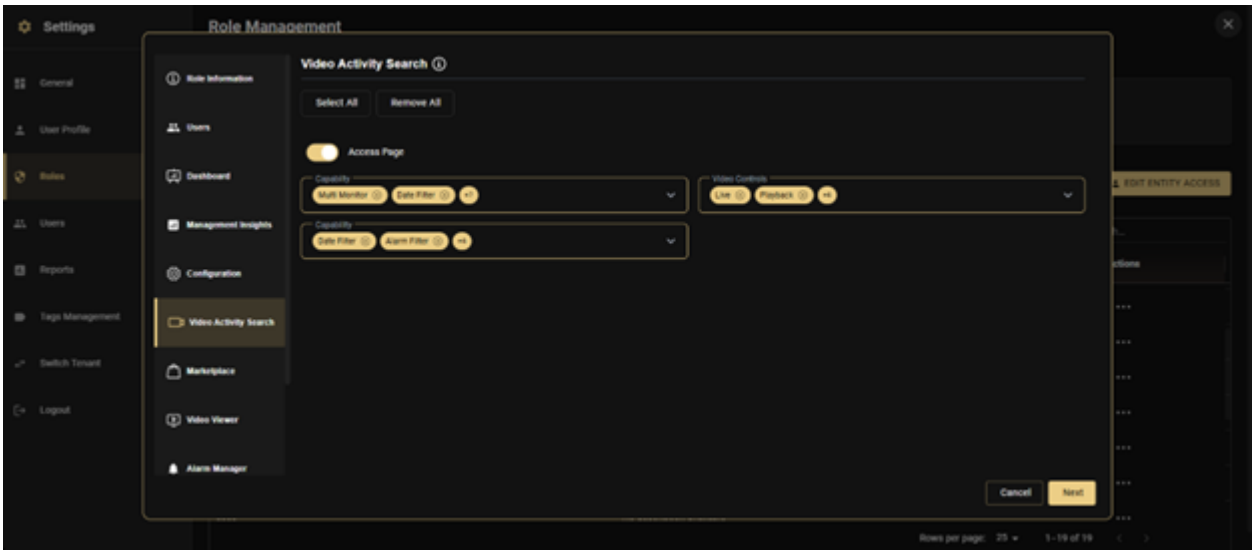
Management Insights:



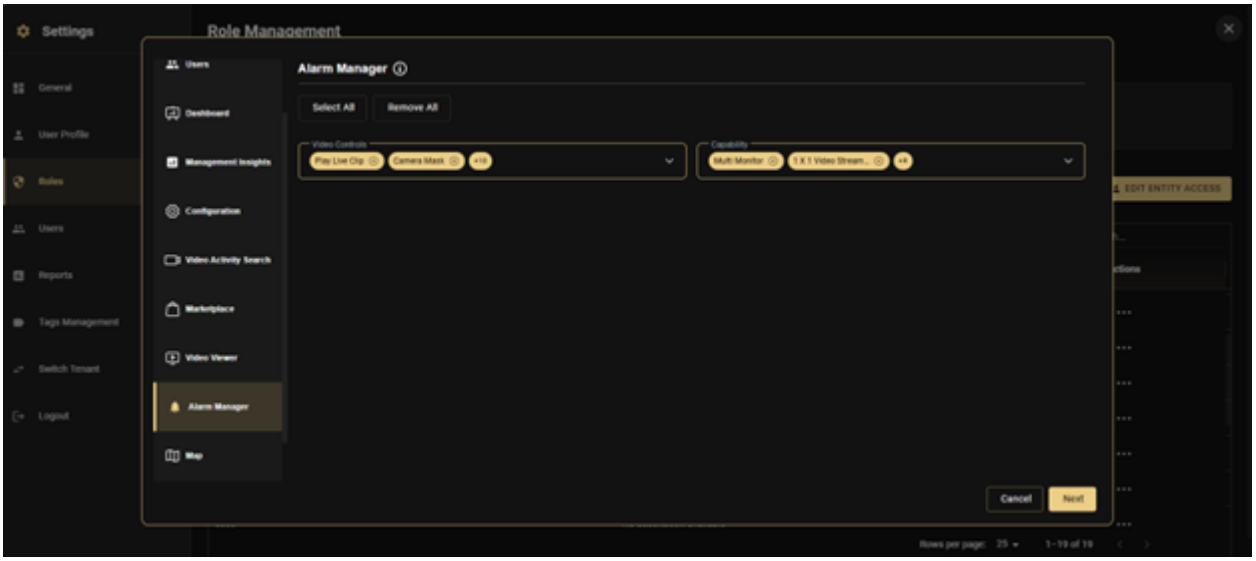
Configuration:



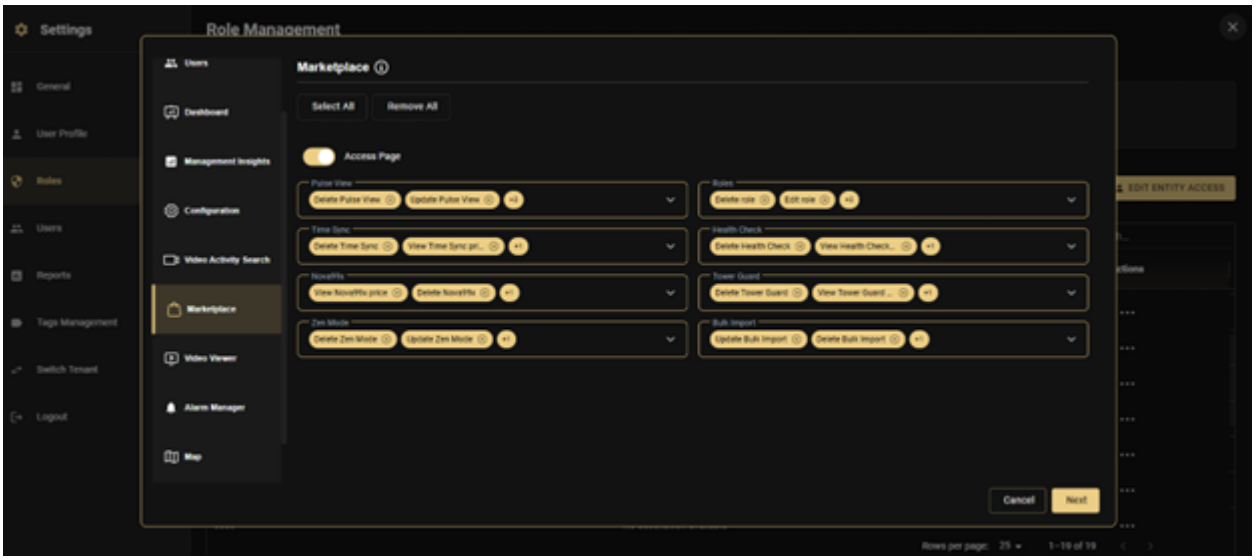
Video Activity Search:



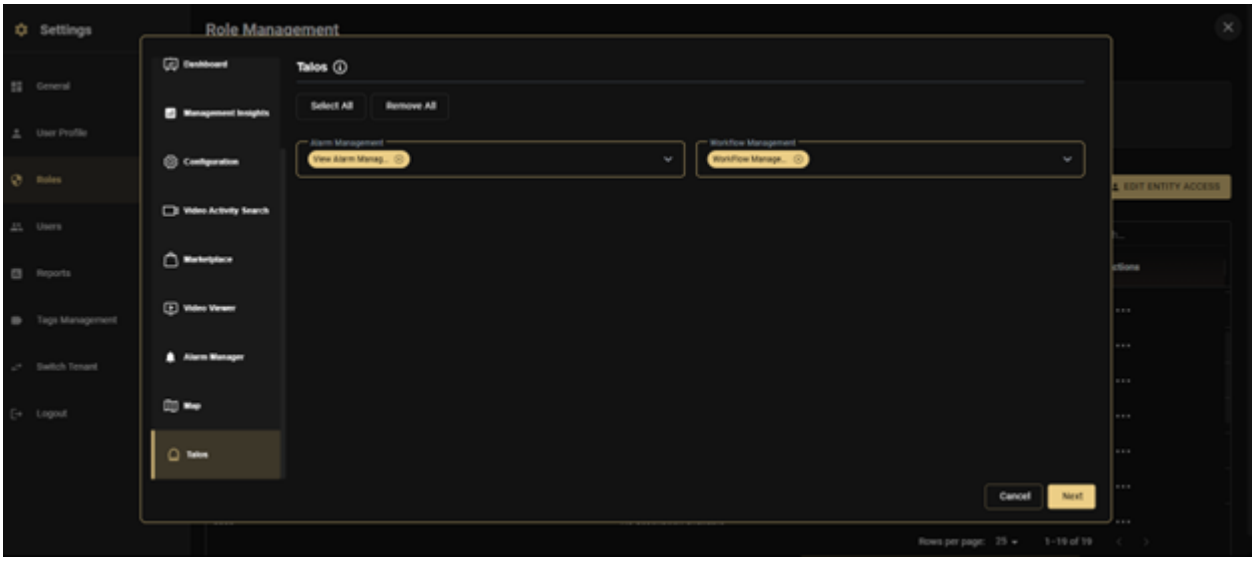
Alarm Manager:



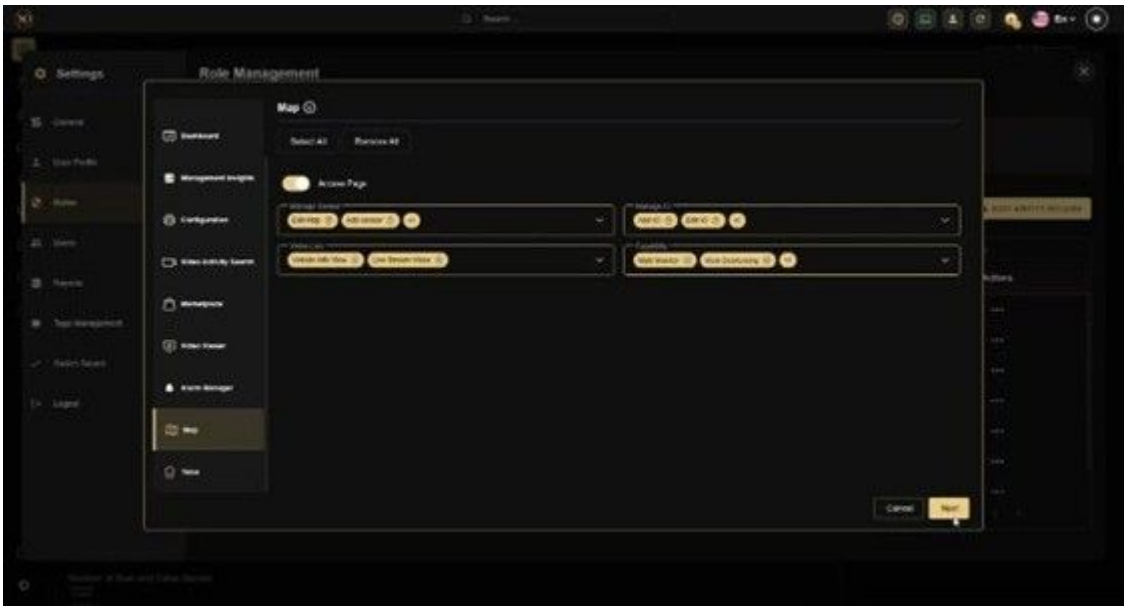
Marketplace:



Talos:



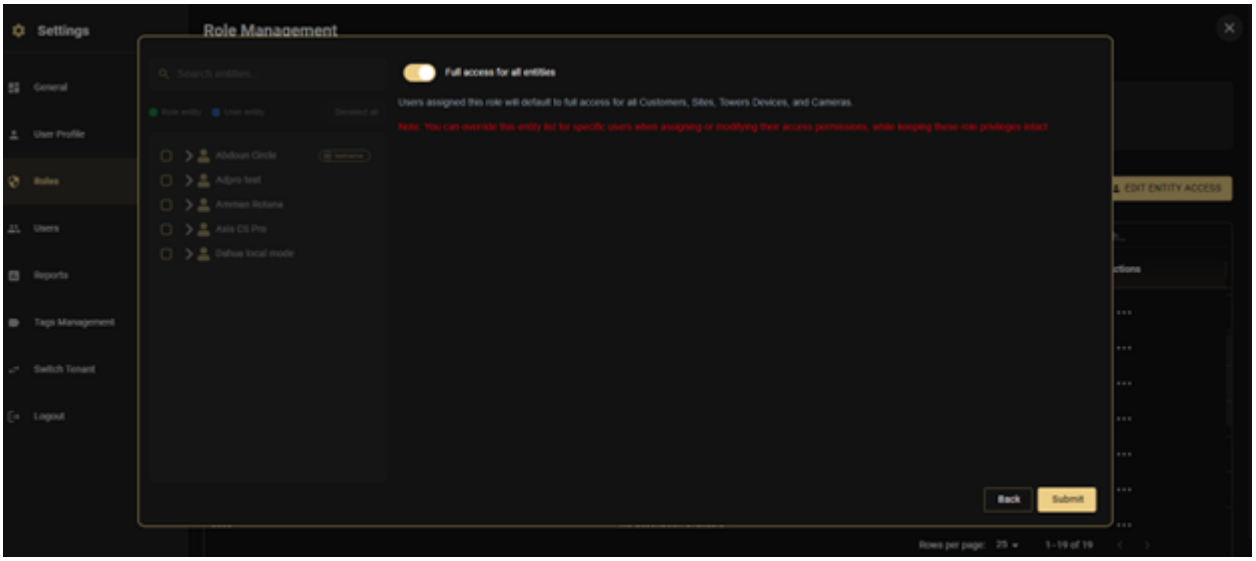
Map:



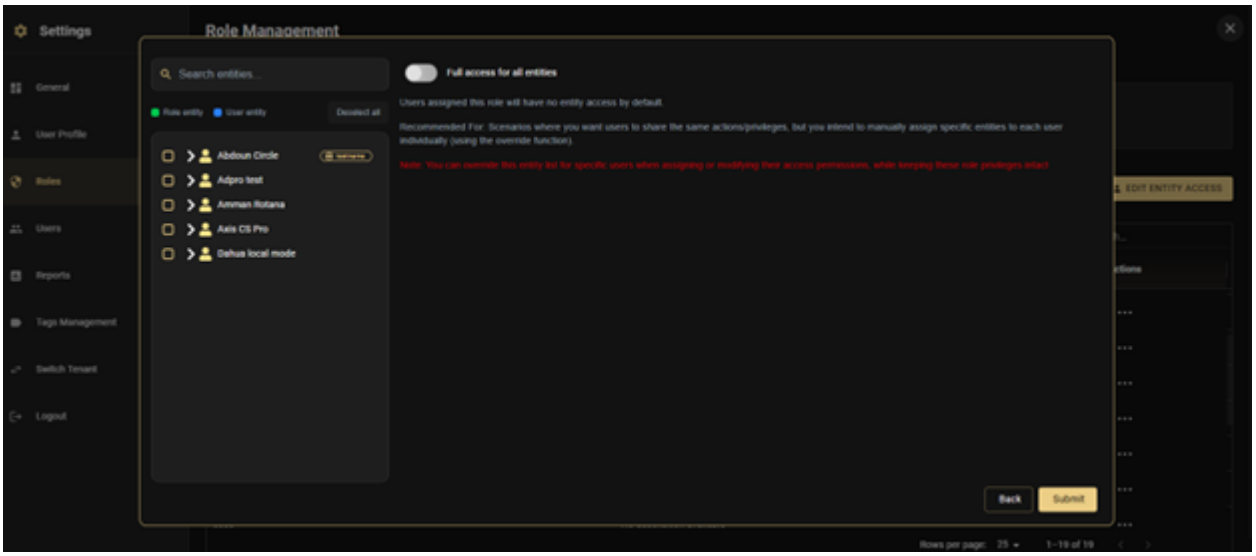
Step 4: Entity Assignment

The final step configures which entities users of this role can access. The entity tree is displayed on the left side with checkboxes for selection and Include Children toggles for each entity.

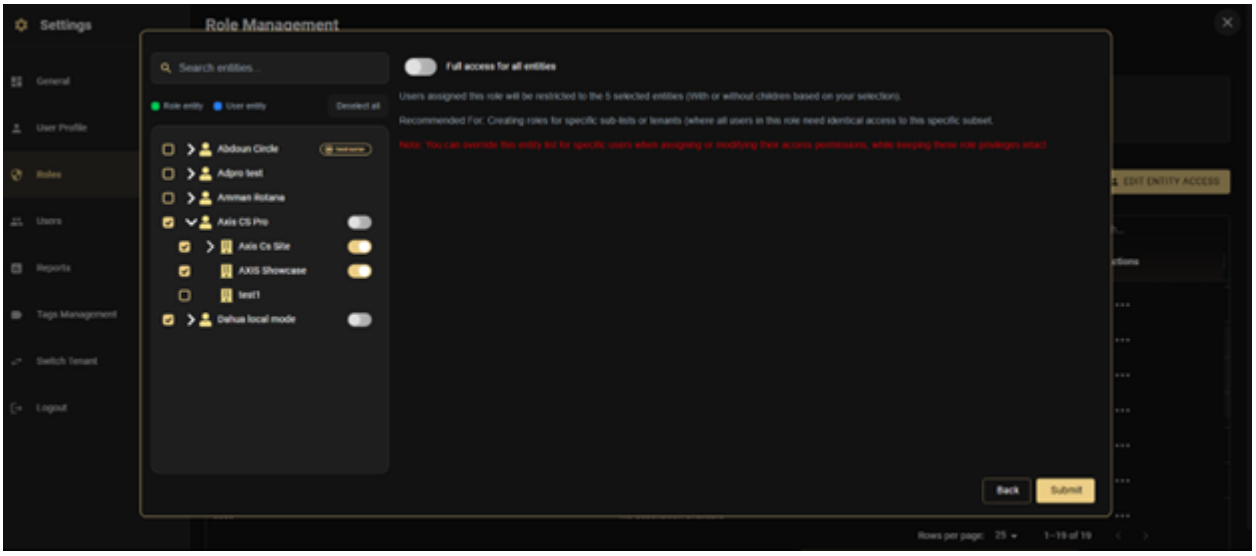
Full Access for All Entities (toggle ON): Users assigned this role will default to full access for all Customers, Sites, Towers/Devices, and Cameras.



No Entity Access (toggle OFF, no entities selected): Users assigned this role will have no entity access by default. Recommended for scenarios where you want users to share the same privileges, but you intend to manually assign specific entities to each user individually using the Edit Entity Access feature.



Selected Entities (toggle OFF, entities checked): Users assigned this role will be restricted to the selected entities. With the Include Children toggle enabled on specific items, all sub-entities (current and future) are automatically included.



Warning: When selecting entities without Include Children, only the currently existing and manually selected items are included. New devices added later will NOT be automatically accessible. Always consider using Include Children to future-proof your entity assignments.

Click Submit to create the role.

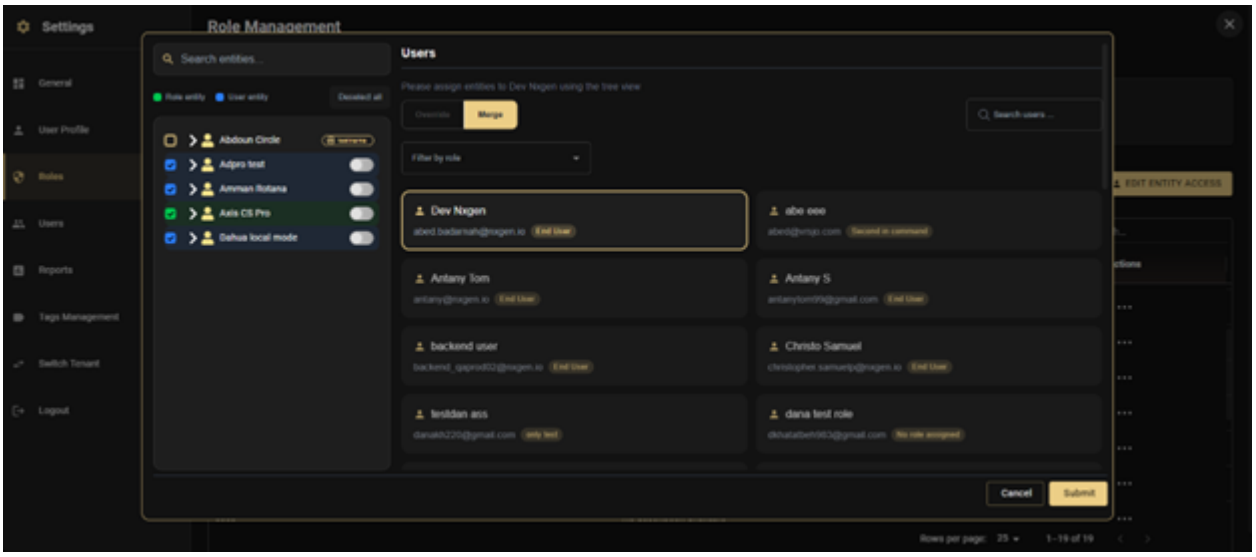
Edit Entity Access (Per-User Entity Management)

Navigation: Settings → Roles → Edit Entity Access

This is the key feature that allows you to customize entity access for individual users without creating separate roles. Remember: entity access determines WHERE a user can operate, not what they can do. Privileges always come from the role.

The interface is split into two panels:

- Left panel: Entity tree with Role Entity, User Entity, and Connected To tabs
- Right panel: Users list with search, role filtering, and Override/Merge mode tabs



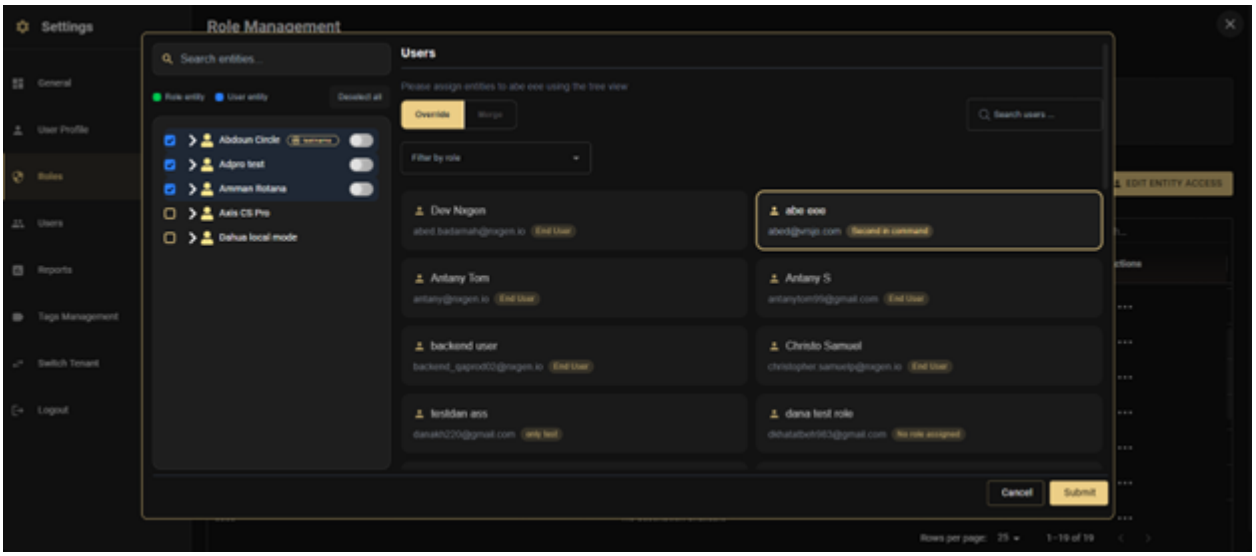
How to Customize a User's Entity Access

1. Click Edit Entity Access from the Role Management page
2. Select a user from the right panel by clicking their card
3. Choose the mode: Override or Merge
4. Use the entity tree on the left to select or deselect entities for that specific user
5. Enable Include Children on entities where you want to include all current and future sub-entities. This is crucial for ensuring that any newly added cameras, devices, or sensors under a parent entity are automatically included in the user's entity access list without requiring manual updates.
6. Click Submit to save changes

Override vs. Merge

Override Mode The user's custom entity selection REPLACES the role-level defaults. Whatever you select for this user overrides what the role provides. Use this when you want complete control over a specific user's entity access.

Merge Mode The selected entities are ADDED on top of the role's existing entity assignments. The user gets both the role-level entities AND the additional user-level entities. Use this to extend a user's access beyond what the role provides.



Tip: Use the "Filter by role" dropdown to quickly find all users assigned to a specific role and customize their entity access in bulk.

Editing a Role

Navigation: Settings → Roles → [Role Name] → Edit (via Actions menu)

The edit flow uses the same wizard interface as role creation. Modify any step — Role Information, Users, Module Privileges, or Entity Assignments — then save. Changes affect all users with this role immediately.

Warning: Removing module access or entities may disrupt user workflows. Verify who has this role before making significant changes.

Deleting a Role

Navigation: Settings → Roles → [Role Name] → Delete (via Actions menu)

The system warns if users are currently assigned to the role and lists affected users. System roles (Admin, Super Admin) cannot be deleted.

Configuration Examples

Example 1: Two Regional Admins, One Role

Scenario: You have two regional admins who need identical privileges but Admin A manages Northeast customers while Admin B manages Southeast customers.

Step 1: Create the role with no default entity access

1. Go to Settings → Roles → Configure New Role
2. Role Information: Name = "Regional Admin", Description = "Full monitoring and alarm control for assigned region"
3. Users: Select both Admin A and Admin B from the user grid
4. Module Privileges: Enable Dashboard, Video Viewer, Alarm Manager, Map with desired capabilities
5. Entity Assignment: Leave "Full access for all entities" OFF and do NOT select any entities. Click Submit.

Step 2: Assign entity access per user

- Go to Settings → Roles → Edit Entity Access
- Select Admin A → Choose Merge → Check all Northeast customers (with Include Children enabled) → Submit
- Select Admin B → Choose Merge → Check all Southeast customers (with Include Children enabled) → Submit

Result: Both admins share the exact same role and privileges, but Admin A only sees Northeast customers while Admin B only sees Southeast customers. One role instead of two.

Example 2: View-Only Operator

Scenario: Operators who should only view cameras during their shift — no video export, no alarm control, just live viewing and playback for specific sites.

- Settings → Roles → Configure New Role
- Role Information: Name = "View-Only Operator"
- Users: Select the operators who need this role
- Module Privileges: Enable only Dashboard (view only) and Video Viewer (with Live View and Playback). Leave all other modules disabled.
- Entity Assignment: Select specific sites these operators cover in the entity tree with Include Children enabled.

Result: Operators can watch cameras but cannot export footage, access configuration, or control alarms.

Example 3: Installer with Site-Specific Access

Scenario: A field installer needs access to a specific customer site to manage devices and configure sensors.

- Create role "Installer" (or use the default Installer role)
- Module Privileges: Enable Configuration (devices, sensors, network settings), Video Viewer (Live View for testing)
- Entity Assignment: No default entity access
- Edit Entity Access: Select the installer user → Merge → Select only the specific site they're working on with Include Children enabled

Result: The installer can only see and manage the specific site and all its devices/sensors. When the job is done, remove their entity access or reassign to the next site.

Example 4: Mobile App Access for Field Technicians

Scenario: Field technicians need mobile app access to sites they service, with camera viewing and alarm control.

- Create role "Field Technician" via the role wizard
- Module Privileges: Enable Video Viewer (Live View), Alarm Manager (Arm/Disarm), Map (Live Stream View)
- Entity Assignment: No default entity access. Use Edit Entity Access to assign each technician their service area with Include Children enabled.

Result: The technician logs into the GCXONE mobile app and sees only their individually assigned sites. They can view live cameras and arm/disarm from their phone. Permission changes apply to mobile within 5 seconds.